Akoizo Original Practice - Form 2 - Answer key

Division C | Mission Possible | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A device ignores the first coin but starts on the second coin. How many start points?

A. 200

B. 0

C. 50

D. 25

Correct: C

2. (2 points) Which platform is legal?

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

A. A flat 10 x 10 x 0.4 cm wooden square with no sides

B. A 12 x 12 cm tray with 1 cm walls

C. A 15 x 15 cm foam pad

D. A cupped 12 cm dish

Correct: A

3. (2 points) What happens if an electrical action fires at 35 seconds?

A. 250-point penalty and that action earns no points

B. 50-point penalty

C. No penalty

D. Tier 3 placement

Correct: A

4. (2 points) A pulley system has four rope segments supporting the load. What is its IMA?

A. 4

B. 1

C. 2

D. 8

Correct: A

5. (4 points) Why can a device with fewer actions outscore one with more?

Unit: Where the points are: building a scoring plan.
Worked-case data: A Regional device starts on the first coin, completes 8 scorable actions and the Final Action at 58 s, earns all ASL points, sets up in 20 minutes, has no adjustments, and measures 60 x 60 x 70 cm. Estimate its score.
Separate investigation observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

answer: Unreliable actions cause jams, touch penalties, loss of the 75-point no-adjustment bonus, wasted time, and possibly no Final Action. (4 points).

6. (4 points) Describe two ways to lengthen a water timer.

Unit: Timing: Target Time, the water timer, and pacing.
Worked-case data: Target Time 60 s. The device runs 67 s with a 25-s water timer that ended at 40 s. Calculate time points and timer bonus.
Separate investigation observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

answer: Use a smaller drain hole to slow flow, or increase the volume that must drain or fill before the next action triggers. (4 points).

7. (4 points) Describe one way a temperature drop can trigger the next action.

Unit: Chemical actions: gas, cold, and fizz.
Worked-case data: You need about 600 mL of CO_2 to inflate a balloon to size. How much baking soda and vinegar should you use?
Separate investigation observations:
Test 1: 1.0 g baking soda + 30 mL vinegar: balloon expanded 14 cm.
Test 2: 2.0 g + 30 mL vinegar: expanded 21 cm.
Test 3: 3.0 g + 30 mL vinegar: expanded 22 cm; undissolved powder left over.

answer: Cooling a sealed container makes the gas contract, pulling a diaphragm or piston that releases a latch. (4 points).

8. (4 points) What happens if the quarter does not start the device?

Unit: The Start Action: coins, slots, and selective triggers.
Worked-case data: Design a rail gap that rejects nickels and dimes but passes quarters to a trigger, allowing 0.5 mm tolerance each side.
Separate investigation observations:
Gap 21.5 mm: Nickel sometimes stuck on the rail and triggered the start.
Gap 22.8 mm: Dime and nickel fell through 20 of 20; quarter carried 20 of 20.
Gap 24.0 mm: Quarter fell through in 6 of 20 trials.

answer: The supervisor gives a 20-second countdown for a team member to drop another quarter from above the device; timing starts with the first action and no start points are awarded. (4 points).

9. (4 points) Why can't one rolling marble trigger two scorable actions?

Unit: The Final Action, the ASL, and competition day.
Worked-case data: Estimate the landing speed of a gum pack dropped 55 cm, and suggest how to keep it on the platform.
Separate investigation observations:
Drop tests: Gum fell 52 cm; landed on platform 7 of 10 times, bouncing off 3.
Change: Added a level release shelf; dropped 51 cm.
New tests: Landed and stayed 10 of 10 times.

answer: Each movable object may be used in only one assigned action; it may start the next action but not be used further. (4 points).

10. (4 points) How can a team prevent room light from falsely triggering the photodiode?

answer: Shield the receiver in a tube aimed at the IR LED, or use an IR filter so only the beam reaches it. (4 points).

11. (6 points) Solve this case and explain your reasoning: A screw has a pitch of 1.5 mm. How many turns move an object 2.0 cm, and is that enough rotation for the action?

reasoning: 2.0 cm = 20 mm.
Turns = 20 mm ÷ 1.5 mm per turn about 13.3 turns.
The rule needs at least 2 full turns and at least 2 cm; 13.3 turns satisfies both. (3 points).

conclusion: A fine-pitch screw easily meets the two-turn rule but needs a long spool string or many turns of a falling weight. (3 points).

12. (2 points) Which plan should the team bring?

Supplied synthetic observations:
Plan A: 12 actions, works in 6 of 10 test runs.
Plan B: 9 actions, works in 10 of 10 test runs.
Failure cost: A failed run usually loses the Final Action, 75 no-adjust bonus, and some time points.

A. Plan B

B. Plan A

C. Neither; add more actions

Correct: A

13. (2 points) About how much water should the timer use?

Supplied synthetic observations:
Timer calibration: 150 mL -> 18 s; 300 mL -> 30 s; 450 mL -> 39 s.
Other actions: Total 46 s without the timer.
Target Time: 80 s announced at State.

A. 150 mL

B. 450 mL

C. About 380 mL

Correct: C

14. (2 points) Define “Limiting reactant” in the context of Mission Possible.

answer: The reactant that runs out first, setting how much product forms. (2 points).

15. (2 points) Define “Coin sorter” in the context of Mission Possible.

answer: Mechanism that separates coins by size or mass. (2 points).

16. (2 points) Define “Platform (tray)” in the context of Mission Possible.

answer: Flat, rigid, unpadded, at most 12 x 12 x 0.5 cm, no sides. (2 points).

17. (2 points) Define “Torricelli’s law” in the context of Mission Possible.

answer: Outflow speed v = sqrt (2gh) from a hole at depth h. (2 points).

18. (2 points) Define “Relay” in the context of Mission Possible.

answer: Electrically operated mechanical switch. (2 points).